Unsupervised Remote Sensing Image Semantic Segmentation Based on Multiscale Contrastive Domain Adaptation

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***Abstract*— Unsupervised domain adaptation (UDA) for remote sensing image semantic segmentation aims to train a deep model on the labeled source domain and apply it to the unlabeled target domain. However, resolution and scene inconsistencies of cross-domain remote sensing images lead to great distribution differences, which weakens the semantic segmentation effect. To solve the above issues, an unsupervised remote sensing image semantic segmentation method is proposed based on multiscale contrastive domain adaptation. First, the mean teacher model is introduced into the UDA paradigm to generate pseudo-labels for target-domain data, thereby achieving the cross-domain segmentation capability. A dynamic class balance sampling (DCBS) method is proposed to mitigate the class imbalance problem in cross-domain data by increasing the sampling frequency of the categories with fewer samples. Then, a data augmentation method called cross-domain mixup (CDMix) is developed to reduce the gap between the source and target domains. Finally, a multiscale cross-domain contrastive loss (MCCL) is developed, which introduces contrastive learning to learn domain-consistent features across the source and target domains, resulting in a more coherent and discriminative feature representation. Experimental results show that the proposed method can yield superior performance for unsupervised remote sensing image semantic segmentation.**

***Index Terms*— Class-balanced sampling, cross-domain contrastive learning, semantic segmentation, unsupervised domain adaptation (UDA).**

I. Introduction

SEMANTIC segmentation involves assigning a class label to every pixel in an image [1], [2], which is a fundamental process in the interpretation of remote sensing images. With advancements in aerial sensor and artificial intelligence technologies, it has become a popular application for satellite image processing [3], such as intelligent traffic [4], agricultural production [5], and land use [6]. In recent years, remarkable achievements have been made in semantic segmentation of remote sensing images driven by advancements in deep learning technology [7], [8], [9], [10], [11], [12]. Nevertheless, these remote sensing foundation models need to be adapted to specific target domains, such as novel geographical locations or different sensor types, which are common in remote sensing applications. In addition, the traditional adaptation process often requires a large number of labeled data from the target domain, which is difficult in many cases [13], [14]. Therefore, the unsupervised domain adaptation (UDA) methods have been proposed to address the domain shift challenge [15], [16], [17], [18]. UDA aims to transfer knowledge from a labeled source domain to an unlabeled target domain, which is essential for enhancing the robustness of remote sensing models in the face of challenges such as sensor variations, climate change, or different imaging modalities. By exploiting UDA, it can improve the adaptability and scalability of remote sensing models for various applications.

The UDA model utilizes the knowledge learned in the source-domain data to perform corresponding tasks in the target domain [19]. The current mainstream UDA methods include adversarial training-based models and self-training-based models. Among them, adversarial training-based models focus on domain differences in cross-domain images, which can be employed for the alignment of cross-domain data across different levels [20], [21]. Yang and Soatto [22] introduced a simple and practical input-level UDA method. The randomly sampled cross-domain images are subjected to extract the corresponding spectral information, and then the low-frequency components are exchanged to minimize the domain discrepancy between cross-domain images. Wang et al. [23] pointed out that performing domain alignment directly from the perspective of the image will ignore the data structure of the underlying categories in the target domain. Thus, they achieved class feature alignment and retained the semantic structure of cross-domain features by a fine-grained domain discriminator. Yang et al. [24] further introduced a context-aware domain adaptive segmentation method employing the cross-attention mechanism, designed to dynamically align domain-invariant features. Tsai et al. [25] explored the UDA method, which designed an adversarial network to conduct spatial domain adaptation across diverse feature levels. These methods can effectively align the feature distributions of the source and target domains, which allows the model to achieve better generalization in the target domain. At the same time, they can dynamically adjust according to the data distribution of the target domain, adapting to different domain variations. However, adversarial training may lead to unstable model training, making it susceptible

Received 5 December 2024; revised 13 March 2025; accepted 9 April 2025. Date of publication 15 April 2025; date of current version 30 April 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 62271396. *(Corresponding author: Jie Geng.)*

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Digital Object Identifier 10.1109/TGRS.2025.3560673

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to adversarial examples and requiring higher computational costs.

Self-training-based UDA models iteratively produce confident pseudo-labels for target-domain data to train the domain adaptation networks [17]. To address the issue of pseudo-label drift, Zou et al. [26] introduced a self-training model grounded in spatial prior knowledge, which refines the generated pseudo-labels. Melas-Kyriazi and Manrai [27] introduced a UDA segmentation model relying on consistent self-training. Tranheden et al. [28] further explored the UDA network to learn the domain robustness features, which employs cross-domain mixed sampling to enhance the robustness of the model. Zhang et al. [29] utilized the prototype distribution of the target domain to estimate pseudo-label confidences, resulting in a more condensed target feature space and achieving significant performance improvements. These methods are simple and easy to implement, usually requiring only fine-tuning on the existing models. They can effectively leverage unlabeled data by generating pseudo-labels for training, thereby improving the model performance. However, the quality of the generated pseudo-labels has a significant impact on model performance. Low-quality pseudo-labels may lead to overfitting and slow convergence. In some cases, adversarial training-based and self-training-based UDA models can be combined to form an ensemble model, which adopts adversarial training for feature alignment and utilizes self-training to enhance data utilization. Additionally, combining these methods enables more comprehensive utilization of both labeled and unlabeled data, which can enhance the overall performance of unsupervised learning.

However, many existing approaches for semantic segmentation based on UDA focus on specific differences in images, such as scene differences or resolution discrepancies. Actually, there are complex domain gaps between cross-domain remote sensing data collected by different sensors in different scenes and times. Additionally, in the UDA baseline, source- and target-domain samples are used to train the student segmentation network by calculating the source-domain supervision loss and target-domain pseudo-loss, where both these losses are cross-entropy losses. However, for UDA tasks, there is a large domain gap between source- and target-domain samples, and there are significant intraclass differences and interclass similarities across different domains. Directly applying cross-entropy loss to cross-domain data may lead to the merging of similar classes, which will result in numerous incorrect pseudo-labels. These incorrect pseudo-labels can make the network more "confident" in the wrong decision boundary through cross-entropy loss, which will seriously affect the performance of segmentation [30]. To address this issue, we employ contrastive learning to learn data representations by increasing the similarity between related samples and decreasing the similarity between unrelated samples.

Therefore, an unsupervised method based on multiscale contrastive domain adaptation is introduced for semantic segmentation, which aims to address the issue of domain differences in resolution and scene for cross-domain remote sensing images. First, a mean model is introduced into the UDA training paradigm, and the dynamic class balance sampling (DCBS) is designed to alleviate the class imbalance issue across cross-domain data. Then, to reduce the domain gap between different data, cross-domain mixup (CDMix) is introduced, where a mixed-domain pseudo-loss with adaptive weights is applied to enhance the robustness. Moreover, a multiscale cross-domain contrastive loss (MCCL) is proposed to learn consistency features between cross-domain data. The primary contributions of this article can be summarized below.

1) An unsupervised semantic segmentation network leveraging multiscale contrastive domain adaptation is proposed, which addresses the domain differences in resolution and scene between cross-domain images. The mean teacher model is introduced into the UDA paradigm and a DCBS is developed, which effectively mitigates the class imbalance problem between the cross-domain data.
2) A data augmentation approach called CDMix is introduced, which reduces the domain gap between cross-domain data as well as improves the robustness of the domain adaption model.
3) An MCCL is proposed to learn the domain consistency features between high-level and low-level features of the cross-domain data.

The remaining sections of this article are organized as follows. Related work is introduced in Section II. Section III provides a detailed introduction to our proposed methodology. Section IV presents experimental validations to assess the performance of the proposed model. Section V summarizes and looks forward to our work.

II. Related Work

A. Remote Sensing Image Semantic Segmentation

Semantic segmentation of remote sensing images is a fundamental and challenging task. In recent years, various deep learning-based methods have been proposed for the semantic segmentation of remote sensing images. For instance, to establish comprehensive relationships among different spatial locations or features, Mou et al. [31] developed a spatial relationship model and a channel relationship model. Moreover, these modules can be easily integrated into existing networks. To alleviate the semantic information loss caused by conventional cropping operations, a two-stage training approach was proposed by Ding et al. [32]. The global semantic features covering extensive areas are acquired in the first stage, while in the second stage, a local feature learning model is built to gather local information from cropped patches. Nogueira et al. [33] trained dilated networks with different patch sizes, enabling the network to model multiscale contextual features from different context patches. During training, the network selects the best patch scale to enhance semantic segmentation accuracy during training.

Recently, due to the higher annotation requirements of supervised learning methods, many researchers have focused on semantic segmentation under weak supervision. Miao et al. [34] proposed a novel method that acquires knowledge from a limited set of labeled data and then leverages pseudo-labels to

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enhance segmentation capabilities. Specifically, a class activation network and an edge-aware network have been proposed to generate more high-confidence pseudo-labels. Lu et al. [35] designed a linear sampling self-training segmentation method, which aims to reduce noise in pseudo-labels by adaptively assigning thresholds to each class. Additionally, research on UDA semantic segmentation methods for remote sensing images has gradually become a hot topic [36], [37], [38], [39]. The UDA utilizes knowledge learned from labeled source-domain data to perform corresponding tasks in the unlabeled target domain, which has significant application prospects.

B. UDA for Semantic Segmentation

Advancements in remote sensing technology have enabled the acquisition of vast amounts of remote sensing data. Nonetheless, significant disparities exist among remote sensing data collected by different sensors across various scenes and time periods. Therefore, remote sensing semantic segmentation based on UDA has become one of the hot topics in recent years.

Wang et al. [40] proposed an unsupervised domain-adaptive semantic segmentation dataset called LoveDA, which is used to test the UDA methods. Xu et al. [15] designed a transferable attention alignment method to address the challenge of the same category but with different characteristics for cross-domain remote sensing images, which can introduce a greater variety of fine-grained features into the network. At the same time, a class-aware adaptive alignment method is developed to model the intraclass compactness and interclass separation of the source and target domains, greatly improving the performance of UDA models. Yan et al. [41] introduced the average teacher model into the UDA paradigm and proposed a cross-average teacher network, where reliable pseudo-labels are assigned to target-domain data and global cross-domain distribution alignment is accomplished. In addition, Tasar et al. [42] proposed a standard adversarial network named Standard-GAN. It is developed to standardize every source-domain and target-domain data to ensure uniform distribution across all cross-domain data. Zhou et al. [43] introduced an UDA semantic segmentation framework that introduced a digital surface model (DSM), which combines remote sensing images and DSMs through a multipath encoder and a multitask decoder and aligns the global data distribution of the cross-domains data through the UDA method. To enhance the recognition of new categories in remote sensing scenes, Li et al. [44] developed a remote sensing knowledge graph (RSKG) to support the inference of unseen scene categories. To infer and recognize unseen categories, Li et al. [45] proposed a zero-shot remote sensing image scene classification (ZSRSSC) paradigm, allowing the model trained on known categories to recognize unseen ones.

Most current researches on remote sensing image semantic segmentation based on UDA focuses on specific differences, such as scene and resolution disparities. However, significant domain gaps exist among cross-domain remote sensing data collected by different sensors across various scenarios and time periods. Therefore, there is an urgent need to develop the UDA methods for semantic segmentation of remote sensing images.

III. Methodology

A. Overall Framework

In this work, an unsupervised remote sensing image semantic segmentation is proposed based on multiscale contrastive domain adaptation, as illustrated in Fig. 1. In our diagram, DAFormer [46] is utilized as the UDA baseline network. The DAFormer introduces the average teacher network into the unsupervised domain adaptive training paradigm. Specifically, labeled source-domain samples are input into the student network for training, while the unlabeled target-domain samples are input into the teacher network to generate pseudo-labels for self-training.

Specifically, considering that there is a large domain gap between cross-domain samples, perturbing only the target-domain data and inputting them into the student network for training can cause network parameters to oscillate, making it difficult to achieve convergence. Therefore, during data processing, CDMix is applied to both source and target-domain samples to generate mixed domain samples. In the loss calculation, CDMix is also used for the target-domain segmentation map and source labels to generate pseudo-labels that encompass mixed domain information. The mixed-domain pseudo-loss, with adaptive weights, is then computed to enhance the robustness of the student network. Simultaneously, considering intraclass differences between features in different domains, an MCCL is proposed. Through MCCL, a more consistent and discriminative high-dimension feature space can be obtained, thus enhancing the segmentation performance of target-domain data even further. Moreover, to address the issue of class imbalance between cross-domain samples, a DCBS is applied to source-domain samples to alleviate long-tail distribution problems.

B. Unsupervised Domain-Adaptive Baseline Network

For the UDA remote sensing semantic segmentation, there is a labeled source-domain dataset $D_S = \{(x_i^S, y_i^S)\}_{i=1}^{N_S}$ and a target-domain dataset $D_T = \{(x_i^T)\}_{i=1}^{N_T}$ without labels. The objective is to use the D_S and D_T to train a domain-robust model, and it can adaptively realize the segmentation task on the target-domain dataset. Among them, x_i^S and y_i^S stand for samples and labels of the source domain, respectively, x_i^T denotes the samples of the target domain, and N_S and N_T are the sample numbers of the source and target domains, respectively. The baseline of the UDA network mainly includes student segmentation network g_θ and teacher segmentation network h_ϕ, where θ and ϕ are parameters of the student and teacher network. The structures of them are the same, and the SegFormer [47] based on Transformer is used in this article.

The student network employs the AdamW optimizer to optimize the network parameters θ during the training process, and the teacher network updates the network parameters ϕ through the θ by EMA as

$$\phi_{(t+1)} = \alpha\phi_t + (1-\alpha)\theta_t. \tag{1}$$

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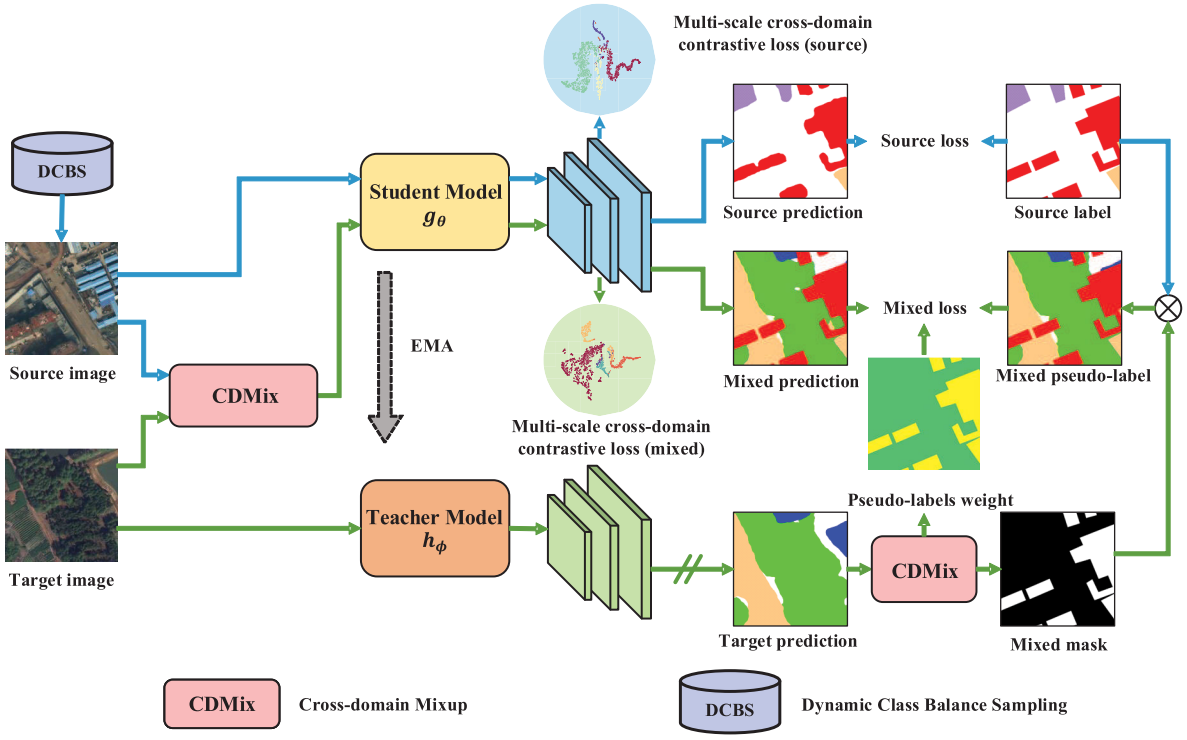


Fig. 1. Multiscale contrastive domain adaptation framework for unsupervised remote sensing semantic segmentation.

Here, EMA stands for the exponential moving average, which is a parameter-sharing method, α and t are the weight of the parameter update and the training epoch, respectively. The source domain D_S only performs supervised learning on the student network and outputs the prediction result $p_i^S = g_\theta(x_i^S)$ and then calculates the source-domain cross-entropy loss with its corresponding label y_i^S as

$$L_S = -\frac{1}{|D_S|}\sum_{(x_i^S, y_i^S)\in D_S}\sum_{j=1}^{H\times W} y_{ij}^S \log\left(g_\theta\left(x_{ij}^S\right)\right) \tag{2}$$

where H and W represent the height and width of the input image, respectively.

For the unlabeled target domain D_T, the teacher network outputs the prediction result $p_i^T = h_\phi(x_i^T)$ first, and then the pseudo-label $\hat{y}_i^T$ can be obtained by

$$\hat{y}_i^T = \operatorname{argmax}\left(h_\phi\left(x_i^T\right)\right). \tag{3}$$

Considering that the generated pseudo-labels may include errors, the pseudo-label confidence weight is defined as

$$q_i^T = \frac{\sum_{j=1}^{H\times W}\left[\max\left(h_\phi\left(x_{ij}^T\right)\right) > \gamma^T\right]}{H \times W} \tag{4}$$

where γ^T is the threshold of the confidence, which is set to 0.95. The pseudo-label confidence weight q_i^T reflects the proportion of high-confidence pixels in the entire data. It is used together with the $\hat{y}_i^T$ to calculate the target-domain pseudo-loss as

$$L_T = -\frac{1}{|D_T|}\sum_{x_i^T\in D_T}\sum_{j=1}^{H\times W} q_{ij}^T \hat{y}_{ij}^T \log\left(h_\phi\left(f\left(x_{ij}^T\right)\right)\right). \tag{5}$$

Here, $f(\cdot)$ is the disturbance function, which refers to color dithering and Gaussian blurring in this article. Then, D_S and D_T are fed into the student network g_θ to learn the cross-domain robustness features. Finally, the parameters h_ϕ are updated through the EMA algorithm to migrate the model trained on the source domain toward the target domain, to achieve the domain adaptation of remote sensing images.

C. Dynamic Class-Balance Sampling

As for cross-domain remote sensing data, due to different scenes, there are problems of class imbalance between cross-domain data. For example, there are more buildings and highways in urban scenes, while more farmland and trees in rural scenes. This imbalance leads the segmentation network to preferentially learn categories with a larger number of samples, thereby biasing the network toward these categories. Because during the training process, the earliest samples will be fit first. Therefore, in the early stage of network training, dynamic sampling of categories with small sample sizes can effectively alleviate the category imbalance problem. Thus, we calculate the frequency of each category in the source domain D_S as

$$f_k = \frac{\sum_{i=1}^{N_S}\sum_{j=1}^{H\times W}\left[y_{ij}^S = k\right]}{N_S \times H \times W} \tag{6}$$

where f_k is the occurrence frequency of the kth category in the entire dataset and $[\cdot]$ denotes the Iverson bracket, when the condition is true, it is 1, otherwise is 0. The sampling probability of the kth category can be calculated as

$$P_k = \frac{e^{(1-f_k)/\alpha_t}}{\sum_{k'=1}^{K} e^{(1-f_{k'})/\alpha_t}} \tag{7}$$

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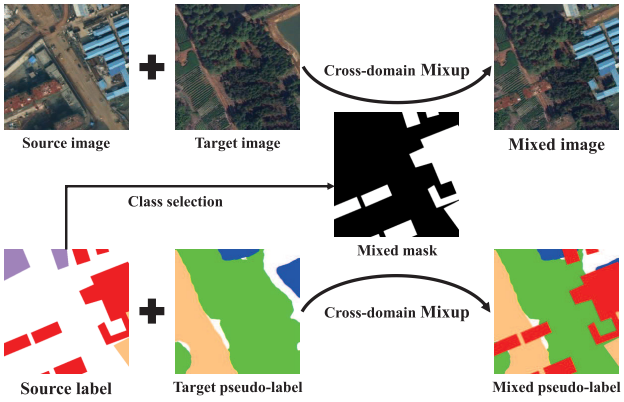


Fig. 2. Example diagram of cross-domain mixup.

where α_t is the probability adjustment coefficient, which is used to adjust the balance of the probability distribution. The larger α_t is, the probability distribution is smoother and it is closer to the sample frequency distribution. On the contrary, the smaller α_t is, the probability distribution is sharper and the sampling probability of rare categories is greater. At the beginning of the training, it is necessary to increase the sampling probability of rare categories to balance the distributions of the dataset. In the later stages of the training, it is crucial to strengthen the capacity to fit common categories in both the source and target domains. Therefore, α_t is dynamically adjusted according to the training progress as

$$\alpha_t = \alpha_0 \cdot \left(1 - \frac{t}{T}\right). \tag{8}$$

Here, t is the current epoch of network training, T is the total epoch, and α_0 is the initial probability adjustment coefficient, which is set to 0.01.

D. Cross-Domain Mixup

In the UDA task, due to the significant domain gap between cross-domain samples, solely employing perturbations on the target-domain samples is insufficient for bridging the domain gap. Moreover, the target-domain samples may contain disturbance noise, which can easily cause parameters of the student network to oscillate and difficult to fit for the target domain. As for remote sensing images, the sample distribution in different domains has the characteristics of local continuity and global discreteness. Thus, a data augmentation method named CDMix is developed, where the processing flow is shown in Fig. 2. We randomly select two images from both the source and target domains for the mixing operation. First, we perform a class selection operation on the source-domain dataset to generate a mixed domain mask M_{ij} as

$$M_{ij} = \begin{cases} 1, & \text{if } y_{ij}^S \in k \\ 0, & \text{otherwise} \end{cases} \tag{9}$$

where y_{ij}^S denotes the source-domain label map and k represents a set of half of the categories randomly selected from the source-domain data. Then, the mixed domain mask is applied for CDMix calculations on source- and target-domain samples to generate mixed domain sample x_i^M as

$$x_i^M = M_i \odot x_i^S + (1 - M_i) \odot x_i^T. \tag{10}$$

Here, x_{ij}^S denotes the source-domain data and x_{ij}^T denotes the target-domain data.

The same method is adopted to generate mixed-domain pseudo-labels $\hat{y}_i^M$. Considering that $\hat{y}_i^T$ is not necessarily accurate, we set the pseudo-label weight W_i for the mixed mask area as

$$W_{ij} = \begin{cases} q_i^T, & \text{if } M_{ij} = 0 \\ 1, & \text{otherwise.} \end{cases} \tag{11}$$

The mixed domain pseudo-loss can be calculated as

$$L_M = -\frac{1}{|D_M|} \sum_{x_i^M \in D_M} \sum_{j=1}^{H \times W} W_{ij} \hat{y}_{ij}^M \log\left(g_\theta\left(x_{ij}^M\right)\right). \tag{12}$$

By using CDMix for source- and target-domain samples, domain-mixed samples can be explicitly generated for the training of the student network. Compared with perturbation processing of target-domain samples, CDMix can learn domain robustness features, as well as reduce the domain gap between the cross-domain data.

E. Multiscale Cross-Domain Contrastive Learning

In UDA, the traditional cross-entropy loss is utilized to train the student network in both the source and target domains. However, there is a large domain gap as well as significant intraclass differences and interclass similarities between the cross-domain samples. The wrong pseudo-labels will make the network more confident about the wrong decision boundaries through cross-entropy loss, which will weaken the segmentation performance. To address this issue, we introduce a contrastive learning method into the UDA network, and an MCCL is proposed in our framework. Specifically, the domain consistency features are learned by contrastive learning between the cross-domain data, as well as between high- and low-level features. This contrastive loss can reduce intraclass differences in cross-domain features and strengthen the distinction between different categories of features, thereby obtaining more consistent and discriminative high-dimensional features.

Specifically, to obtain domain consistency features of the cross-domain data, four multiscale features $\mathbf{F} = \{F_l, l \in [1, 4]\}$ of the encoder are extracted, and they are fused in the decoder to obtain the fused feature F. It leverages the hierarchical structure of the encoder, where each stage captures different details from the data. Here, F^S and F^M represent the fused features obtained on the source domain and mixed domain, respectively. For MCCL, we introduce the pixel-level contrastive learning loss InfoNCE, which can be defined as follows:

$$L_{\text{NCE}} = -\frac{1}{K \times |Q_i|} \sum_{k=1}^{K} \sum_{j=1}^{H \times W} \sum_{v_{ijk} \in Q_i} \times \left[\log \left(\frac{e^{\langle v_{ijk}, v_{ik}^+ \rangle / \tau}}{e^{\langle v_{ijk}, v_{ik}^+ \rangle / \tau} + \sum_{v_{ijk}^- \in N_i} e^{\langle v_{ijk}, v_{ijk}^- \rangle / \tau}} \right) \right] \tag{13}$$

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where Q_i represents the query feature vector of the ith sample, N_i represents the negative sample set of query feature vectors, v_{ik} represents the query feature vector of the kth category in Q_i, and v_{ik}^+ and v_{ik}^- represent the positive and negative samples of the query feature vector, respectively. $\langle \cdot, \cdot \rangle$ represents the cosine similarity and τ is the temperature coefficient, which is set to 0.5. This contrastive loss calculates the similarity between positive and negative samples, which can compare the feature embedding of different categories to obtain more discriminative high-dimensional features. The essence of this loss lies in effectively optimizing features of both the positive and negative sets. Considering that the source domain has labels but the target domain has no labels, we define the positive and negative samples for the source and target domains, respectively.

For the source-domain fused feature $F^S \in \mathbb{R}^{H\times W\times d}$, we first vectorize it into $V^S \in \mathbb{R}^{\mathrm{HW}\times d}$, which means it contains HW feature vectors with length d. Then, the query feature vector set is defined as

$$Q_i^S = \left\{ v_{ijk} | y_{ij}^S = k,\, p_{ij}^S(k) > \delta^S \right\}_{k=1}^K \tag{14}$$

where δ^S represents the threshold of the positive sample and v_{ijk} represents the ith query feature vector of the jth sample, which satisfies that the predicted probability is greater than δ^S. For the query feature vector of the same category, its positive sample is unique and can be defined as

$$v_{ik}^+ = \frac{1}{\left|Q_i^S\right|} \sum_{v_{ijk}\in Q_{ik}^S} v_{ijk}. \tag{15}$$

For the query feature vector of the source domain with the kth category, its negative sample set is defined as

$$N_i^S = \left\{ v_{ijk}^- | \left[y_{ij}^S \neq k \right] \cdot \left[0 \le s\left(p_{ij}^S(k) \right) \le \lambda_l \right] \right\}_{k=1}^K \tag{16}$$

where $s(\cdot)$ denotes the sorting function and $\lambda_l = 3$ is the low-level sorting threshold. Through contrastive learning of positive and negative samples, the discrimination of interclasses features can be enhanced.

For the mixed domain fused feature $F^M \in \mathbb{R}^{H\times W\times d}$, it is also vectorized into $V^M \in \mathbb{R}^{\mathrm{HW}\times d}$. Since it is unlabeled, it can generate the corresponding pseudo-label $\hat{y}_{ij}^M$, and then the mixed domain query feature set is defined as

$$Q_i^M = \left\{ v_{ijk} | \hat{y}_{ij}^M = k,\, p_{ij}^M(k) > \delta^S \right\}_{k=1}^K. \tag{17}$$

Similarly, the positive sample of the same category is unique, which is defined as

$$v_{ik}^+ = \frac{1}{\left|Q_i^M\right|} \sum_{v_{ijk}\in Q_{ik}^M} v_{ijk}. \tag{18}$$

For the mixed-domain query feature vector, since it is unlabeled and the generated pseudo-label may be inaccurate, its negative sample set can be defined as

$$N_i^M = \left\{ v_{ijk}^- | \left[p_{ij}^S < \gamma^M \right] \cdot \left[\lambda_l \le s\left(p_{ij}^M(k) \right) \le \lambda_h \right] \right\}_{k=1}^K \tag{19}$$

where γ^M is the confidence threshold, and λ_h is the high-level sorting threshold.

Finally, the source-domain multiscale contrastive loss L_{NCE}^S and the mixed domain multiscale contrastive loss L_{NCE}^M can be obtained through (13). At the same time, combining (2) and (12), the overall loss function of our proposed method can be obtained as follows:

$$L_{\mathrm{total}} = L_S + L_M + L_{\mathrm{NCE}}^S + L_{\mathrm{NCE}}^M. \tag{20}$$

IV. Experiments

A. Datasets

In our work, ISPRS and LoveDA datasets are utilized for experimental analyses. The ISPRS dataset is a semantic annotation dataset released by the International Society for Photogrammetry and Remote Sensing (ISPRS), including Potsdam and Vaihingen datasets. It covers six typical ground objects, namely impermeable surfaces, buildings, low vegetation, trees, cars, and backgrounds. The Potsdam includes 38 remote sensing images of the size 6000 × 6000 with an image resolution of 5 cm. The Vaihingen includes 33 remote sensing images with a size of about 2500 × 2000 and an image resolution of 9 cm. In the experiments, the number of images used for training, testing, and validation sets are 19, 14, and 5 for the Potsdam dataset and 11, 17, and 5 for the Vaihingen dataset, respectively.

LoveDA [40] is a land-cover semantic segmentation dataset, including Urban and Rural two subdatasets. It contains seven typical land objects, namely background, buildings, roads, water, barren, forest, and farmland. It contains a total of 5987 remote sensing images of size 1024 × 1024 with a resolution of 0.3 m.

B. Implementation Details

To validate the effectiveness of the proposed multiscale contrastive domain adaptation, experimental verification is conducted on the ISPRS and LoveDA datasets. These two subdatasets of the ISPRS dataset have different resolutions and sensors. To conduct UDA, domain adaptation experiments from Potsdam to Vaihingen (Potsdam → Vaihingen) and Vaihingen to Potsdam (Vaihingen → Potsdam) are operated, respectively. For the LoveDA dataset, the Urban subdataset mainly acquires the urban areas, while the Rural subdataset captures the rural areas, so it is mainly oriented to domain adaptation tasks with different scenarios. Similarly, domain adaptation experiments from Urban to Rural (Urban → Rural) and from Rural to Urban (Rural → Urban) are conducted to prove the effectiveness of the proposed method.

In the training process of the proposed method, the student network uses AdamW as the optimizer with a weight decay of 0.01, and the momentum is 0.999. Furthermore, the initial learning rate is 6×10^{-4} with a decay ratio of 0.9. The sliding average coefficient of parameter update for the teacher network is 0.999. The proposed method has gone through a total of 50 training epochs and obtained the model with the best performance on the validation set. Finally, we evaluate the performance of our proposed model using OA, mIoU, and $F1$ score [2].

C. Ablation Study

To validate the effectiveness of the DCBS, CDMix and MCCL, ablation experiments are performed under the

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TABLE I
COMPARISONS OF ABLATION EXPERIMENTS ON POTSDAM → VAIHINGEN

Model	Module (✓ stands for adoption)			Overall Metrics		
	DCBS	**CDMix**	**MCCL**	mIoU	mF1	OA
1				53.45	69.83	72.01
2	✓			57.57	71.58	80.23
3	✓	✓		60.12	73.79	81.06
4	✓		✓	62.09	75.42	81.61
5	✓	✓	✓	**64.73**	**77.80**	**82.03**

TABLE II
COMPARISONS OF ABLATION EXPERIMENTS ON RURAL → URBAN

Model	Module (✓ stands for adoption)			Overall Metrics		
	DCBS	**CDMix**	**MCCL**	mIoU	mF1	OA
1				45.27	61.38	60.56
2	✓			48.60	65.00	63.42
3	✓	✓		50.73	66.82	65.00
4	✓		✓	49.86	66.15	64.39
5	✓	✓	✓	**51.92**	**67.95**	**65.96**

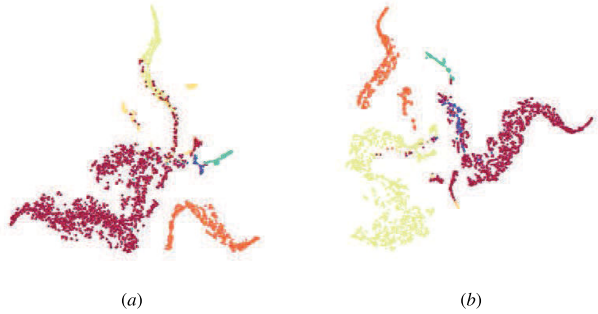
cross-domain conditions of Potsdam → Vaihingen and Rural → Urban.

Table I shows the ablation experimental results under the Potsdam → Vaihingen domain adaptation condition. As observed in the table, DCBS leads to significant improvements in mIoU, mF1, and OA, indicating that the Potsdam dataset does have a serious class imbalance problem. At the same time, when CDMix and MCCL are added, the segmentation effects are also significantly improved. To further demonstrate the robustness of the proposed method, ablation experiments are also conducted under the Rural → Urban domain-adaptive conditions, as shown in Table II. Similarly, when adding each module we proposed, there is a certain degree of performance improvement. According to the experimental results, it is evident that our method has good adaptability with inconsistent domain distribution and resolution or different scenes.

To express the segmentation capabilities of the proposed method more intuitively, the t-SNE algorithm [53] is utilized to obtain visualization results of the high-dimensional features from both the baseline [46] and our proposed method. As shown in Fig. 3, the baseline can generate separated features, but the boundaries between different categories are not obvious, especially since the mixed phenomenon exists in categories with fewer samples. The proposed method can strengthen the feature distinction between different categories, and the visualization results also show more consistent and discriminative high-dimensional features, which is conducive to domain adaption learning on target-domain data.

D. Comparison Experiments

1) Results on the ISPRS Data: To demonstrate the effectiveness of our method, we conduct comparative experiments on the ISPRS dataset. The comparative methods mainly include the baseline DAFormer [46], the most advanced unsupervised segmentation methods including LSR [48], MIC [49], CaGAN [15], and the latest methods including UemDA [50], JDAF [51] and De-GLGAN [52].



(*a*) (*b*)

Fig. 3. T-SNE visualization results of high-dimensional features. (a) Baseline [46]. (b) Our method.

Experimental comparison results under the conditions of Potsdam → Vaihingen are shown in Table III. In addition, we also show the experimental results without domain adaptation (source only), that is, evaluate the model trained solely on the source-domain data and using the target-domain data for testing. As observed in the table, our proposed method achieves 64.73%, 77.80%, and 82.03%, respectively, in the three evaluation indicators of mIoU, mF1, and OA, all of which achieve state-of-the-art performance. Compared with the model without domain adaptation, the proposed method has improved by 12.84%, 10.72%, and 12.76% in the mIoU, mF1, and OA, respectively. At the same time, it is observed that our proposed method yields higher accuracies than UemDA, JDAF, and De-GLGAN. The above analyses illustrate that our method can optimally utilize cross-domain data to minimize the domain gap. In addition, the IoU of each class also demonstrates our method yields superior performance on impervious surfaces, buildings, and trees.

Additionally, we select three visual results of representative test samples to illustrate the superiority of our method, which are shown in Fig. 4. The lower right corner of the first row in the figure is a dense residential area with trees and impervious surfaces. It can be seen that there is an obvious overfitting phenomenon in the building when directly using the source-domain model. However, the proposed method uses UDA to reduce the distribution differences of cross-domain data and can effectively address the issue of overfitting on the source domain. In addition, our proposed method can achieve more refined semantic segmentation results through contrastive learning of multiscale cross-domain features compared with the baseline.

Compared results under the condition of Vaihingen → Potsdam are shown in Table IV. The results are basically consistent with the results under the condition Potsdam → Vaihingen. The proposed method delivers superior segmentation performance in IoU of each category, overall mIoU, mF1, and OA. In addition, the visualization results under this condition are shown in Fig. 5, where three representative results are displayed. The figure illustrates that the model

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TABLE III
COMPARATIVE RESULTS UNDER POTSDAM → VAIHINGEN CONDITION

Methods	Per-class IoU (%)					Overall Metrics (%)		
	Impv.	Building	Low.	Tree	Car	mIoU	mF1	OA
Source only	59.39	69.07	34.34	61.91	34.76	51.89	67.08	69.27
DAFormer [46]	68.85	79.35	38.16	63.84	37.63	57.57	71.58	80.23
LSR [48]	57.11	70.24	28.73	56.55	23.65	47.26	53.56	67.65
CaGAN [15]	63.71	72.81	33.97	54.79	33.91	51.84	-	71.57
SPGAN-DA [13]	67.88	82.83	47.27	55.26	49.79	60.61	-	-
DSM-Assisted [43]	60.75	70.10	22.78	51.81	38.21	57.92	63.73	-
MIC [49]	72.46	77.80	50.05	59.69	**54.68**	62.93	76.74	78.56
UemDA [50]	69.97	80.43	**53.90**	65.63	35.87	61.16	-	-
JDAF [51]	68.76	77.19	47.39	53.38	42.76	57.89	-	-
De-GLGAN [52]	65.34	79.66	42.19	65.04	41.15	58.67	73.55	78.52
Ours	**73.62**	**82.90**	47.27	**67.38**	52.49	**64.73**	**77.80**	**82.03**

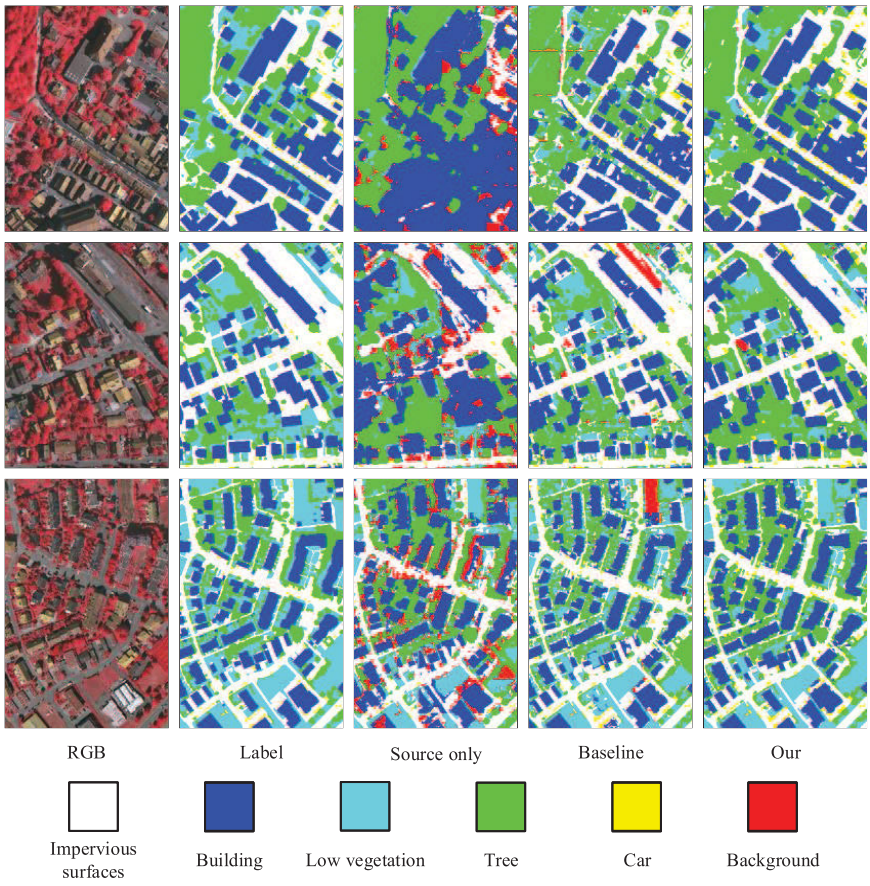


Fig. 4. Visualization results on the test data under Potsdam → Vaihingen condition.

without domain adaptation has basically lost the ability to distinguish low-vegetation and trees, and there is a serious overfitting phenomenon on low-vegetation. At the same time, although the baseline can alleviate the overfitting problem to a certain extent, there is an obvious segmentation discontinuity phenomenon. The above results show that our method effectively mitigates the class imbalance issue across cross-domain data, adaptively learning the domain consistency features of cross-domain data. In other words, our method can reduce the intraclass difference of cross-domain features, thereby improving the segmentation performance of target domain samples.

2) Results on the LoveDA Data: To prove the robustness of our method, comparative experiments are also conducted on the LoveDA dataset. The comparative methods include the baseline DAFormer [46], the most advanced

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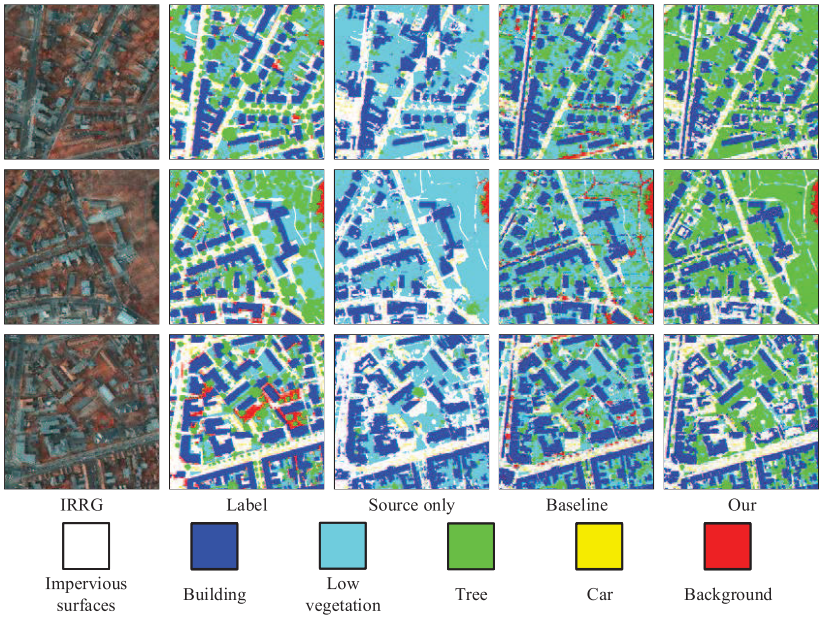


Fig. 5. Visualization results of the test data under Vaihingen → Potsdam condition.

TABLE IV
COMPARATIVE RESULTS UNDER VAIHINGEN → POTSDAM CONDITION

Methods	Per-class IoU (%)					Overall Metrics (%)		
	Impv.	Building	Low.	Tree	Car	mIoU	mF1	OA
Source only	63.40	67.27	45.86	25.29	40.54	48.47	63.34	66.12
DAFormer [46]	57.26	70.83	48.37	37.69	54.82	53.79	67.38	71.16
LSR [48]	47.47	63.19	42.57	30.93	39.42	44.72	51.01	63.01
CaGAN [15]	56.38	50.76	37.15	37.13	32.81	42.85	-	70.75
DSM-Assisted [43]	59.54	68.16	20.96	34.73	49.40	46.56	61.61	-
MIC [49]	64.96	**79.43**	10.33	38.70	45.08	47.70	60.79	68.32
UemDA [50]	57.52	67.71	**55.16**	40.30	56.61	55.46	-	-
JDAF [51]	67.70	76.36	51.59	36.21	**63.22**	59.01	-	-
De-GLGAN [52]	42.20	61.76	50.31	**42.86**	56.63	50.75	56.61	65.73
Ours	**68.27**	77.88	51.24	42.52	60.24	**60.03**	**72.36**	**79.26**

unsupervised semantic segmentation methods including IAST [54], MIC [49], DCA [39], and the latest methods including UemDA [50], JDAF [51], and De-GLGAN [52]. Compared with the ISPRS dataset, the LoveDA has a larger image size but smaller resolution, and the source and target domains have a more serious class imbalance problem. Therefore, the UDA task on the LoveDA dataset is more difficult and challenging.

Comparison results under Rural → Urban condition are shown in Table V. The table shows that compared with other mainstream methods, the proposed method achieves 51.92%, 67.95%, and 65.96% in three evaluation indices, respectively, all achieving the best performance. By comparing with the experiments without domain adaptation (source only), our method improves mIoU, mF1, and OA by 9.72%, 10.46%, and 13.53%, respectively. The results demonstrate that our proposed method effectively addresses the challenges posed by diverse scenarios in cross-domain datasets, and reduces the domain gap between the cross-domain data.

Moreover, the visualization results under this experimental condition are shown in Fig. 6. It shows that the model without domain adaptation has an overfitting problem in building categories. Especially for dense building areas, the overall segmentation result map is rougher and inaccurate. The segmentation results of our method exhibit enhanced detail and accuracy, and correct identification can also be achieved in dense areas of similar categories, which validates the effectiveness of our method.

Comparison results under the Urban → Rural condition are shown in Table VI. Similarly, the proposed method achieves the best performance in three evaluation indicators, reaching 46.63%, 61.13%, and 69.24%, respectively. For the IoU of

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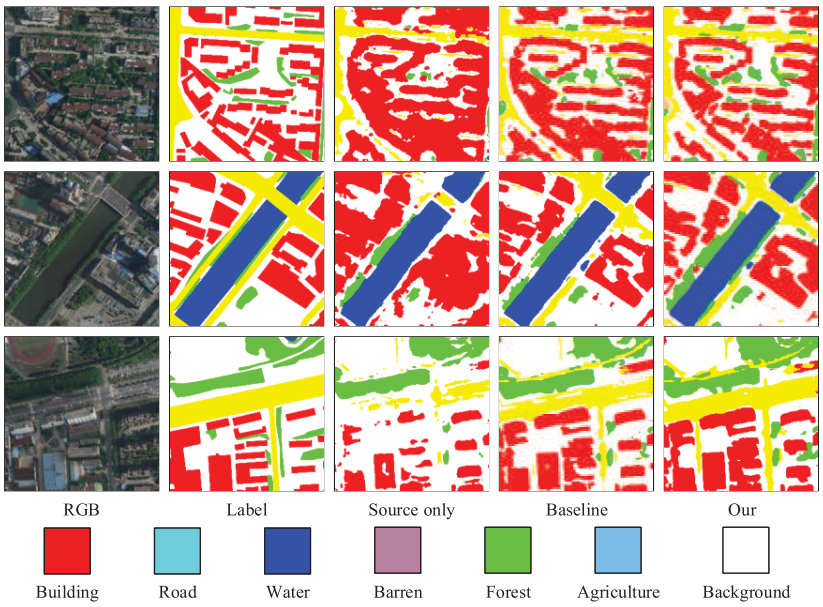


Fig. 6. Visualization results of the test data under Rural → Urban condition.

TABLE V
COMPARATIVE RESULTS UNDER RURAL → URBAN CONDITION

Methods	Per-class IoU (%)							Overall Metrics (%)		
	Back.	Building	Road	Water	Barren	Forest	Agri.	mIoU	mF1	OA
Source only	35.51	55.29	47.47	60.85	43.76	10.09	42.41	42.20	57.49	58.43
DAFormer [46]	35.15	59.65	49.71	55.99	39.10	48.88	51.74	48.60	65.00	63.42
IAST [54]	48.57	31.51	28.73	**86.01**	20.29	31.77	36.50	40.48	-	-
DCA [39]	45.82	49.60	51.65	80.88	16.70	42.93	36.92	46.36	-	-
SPGAN-DA [13]	35.45	50.31	51.02	68.42	42.68	40.17	51.90	48.56	-	-
MIC [49]	39.76	44.03	56.05	54.21	43.56	46.63	41.05	46.47	63.62	61.20
UemDA [50]	46.91	48.28	49.40	83.89	15.58	41.26	34.29	45.66	-	-
JDAF [51]	**49.75**	42.62	39.55	70.79	36.12	39.71	51.29	47.12	-	-
De-GLGAN [52]	35.58	30.99	43.12	68.91	41.56	43.44	33.97	42.51	56.82	53.06
Ours	37.05	**61.79**	**56.15**	60.54	**44.28**	**50.45**	**53.19**	**51.92**	**67.95**	**65.96**

every category, our method achieves the best performance on buildings, roads, and barren. Finally, the visualization results are shown in Fig. 7. It can be clearly seen in the third row of results that the model without domain adaptation has almost lost the recognition ability in the wasteland and forest categories, and the baseline method can only identify some wasteland and forest categories with obvious characteristics. However, our method generates mixed domain samples through CDMix to facilitate the network in acquiring domain robust features, which can alleviate the category imbalance problem between cross-domain data, thereby having effective identification of wasteland and forest.

Through experimental analyses of two datasets, the proposed method can overcome the distribution difference between the cross-domain data, which can enhance the semantic segmentation results under unsupervised conditions.

E. Hyperparameters and Computational Complexity Analysis

To demonstrate the impact of different hyperparameters on the model performance, this section conducts a specific analysis. First, various hyperparameters involved in the proposed model are kept consistent with the baseline network DAFormer [46], including optimizer parameters, the learning rate, the moving average coefficient, and the confidence threshold. Therefore, we only need to discuss the initial probability adjustment coefficient α_0 in the DCBS, as well as the ranking thresholds λ_l and λ_h in MCCL. Since the sampling probability of rare classes needs to be increased during the early stages of training, a smaller value of α_0 is required. Therefore, α_0 is set to 0.001, 0.01, 0.1, 0.2, and 0.5 for discussions. As for the ranking thresholds λ_l and λ_h, since they represent the probability ranking of sample classes, the threshold λ_l is set to 1, 2, and 3, and the threshold λ_h is set to 5, 6, and 7 for

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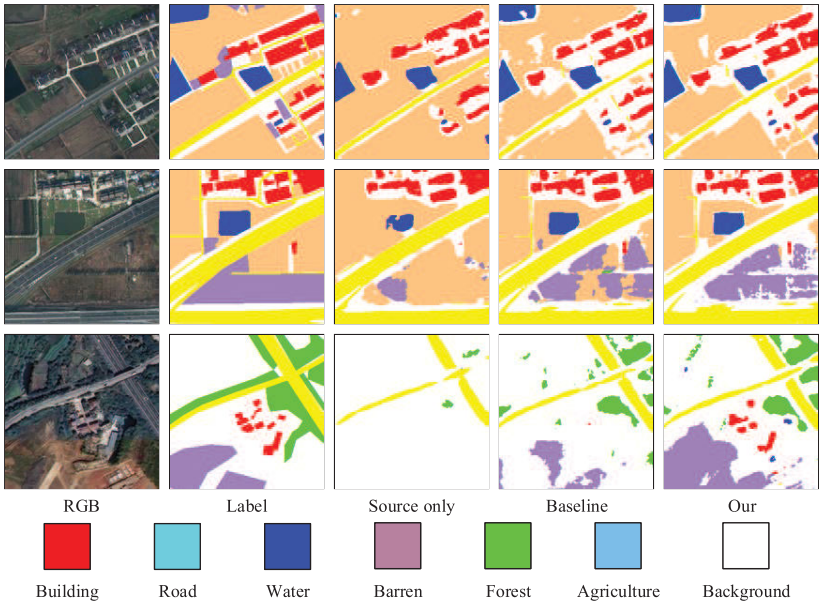


Fig. 7. Visualization results of the test data under Urban → Rural condition.

TABLE VI
COMPARATIVE RESULTS UNDER URBAN → RURAL CONDITION

Methods	Per-class IoU (%)							Overall Metrics (%)		
	Back.	Building	Road	Water	Barren	Forest	Agri.	mIoU	mF1	OA
Source only	53.00	47.27	35.89	51.92	5.79	3.78	38.13	33.68	46.87	47.66
DAFormer [46]	55.02	52.15	37.91	63.17	7.25	28.97	44.81	41.33	56.04	66.85
IAST [54]	29.97	49.48	28.29	64.49	2.13	33.36	**61.37**	38.44	-	-
DCA [39]	36.38	55.89	40.46	62.03	22.01	**38.92**	60.52	45.17	-	-
SPGAN-DA [13]	55.06	50.71	33.80	65.01	9.07	25.43	55.44	42.07	-	-
MIC [49]	47.14	53.01	51.67	64.66	6.64	14.66	43.18	40.14	54.05	63.10
UemDA [50]	30.54	52.42	33.07	31.09	24.75	13.67	56.87	34.63	-	-
JDAF [51]	48.82	43.11	45.24	50.40	27.33	10.02	58.23	40.45	-	-
De-GLGAN [52]	**56.64**	55.99	50.25	**68.25**	10.02	12.86	47.56	43.08	56.57	68.44
Ours	41.50	**56.00**	**65.20**	38.46	**68.78**	11.49	52.87	**46.63**	**61.13**	**69.24**

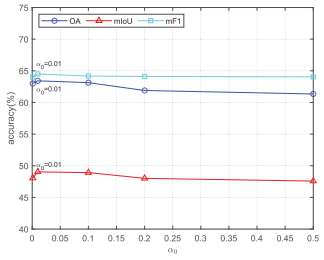


Fig. 8. Sensitivity analyses of hyperparameter α_0 under Rural → Urban domain adaptation conditions.

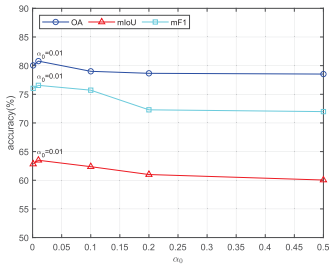
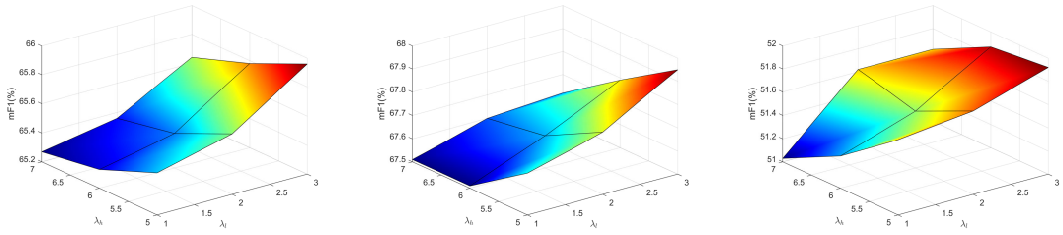


Fig. 9. Sensitivity analyses of hyperparameter α_0 under Potsdam → Vaihingen domain adaptation conditions.

comparisons and discussions. During the experiments, we first find the optimal α_0 and then find the ranking threshold λ_l and λ_h. Experiments have been conducted under the cross-domain condition of Rural → Urban and Potsdam → Vaihingen, the results of the parameter discussions are graphed in the following Figs. 8–11.

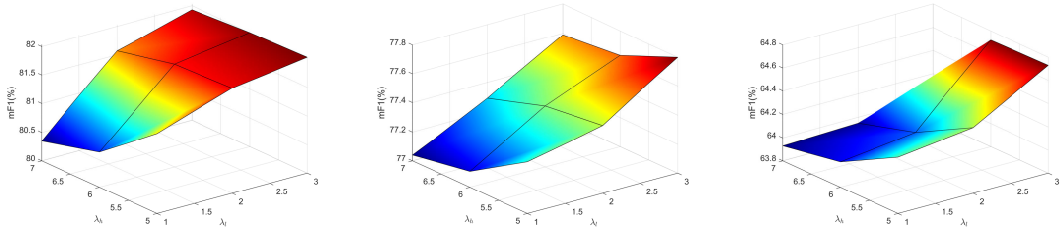
From Figs. 8 and 9, it can be seen that when the initialization probability adjustment coefficient α_0 is set to 0.01, the

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(a) (b) (c)

Fig. 10. Sensitivity analyses of hyperparameters λ_l and λ_h under Rural → Urban domain adaptation conditions. (a) OA. (b) mF1. (c) mIoU.



(a) (b) (c)

Fig. 11. Sensitivity analyses of hyperparameters λ_l and λ_h under Potsdam → Vaihingen domain adaptation conditions. (a) OA. (b) mF1. (c) mIoU.

TABLE VII
ANALYSES OF THE COMPUTATIONAL COMPLEXITY

Methods	DAFormer	MIC	Ours
Params (M)	85.15	85.68	97.23
FLOPs (G)	92.47	83.59	123.46

semantic segmentation accuracies achieve the best on each dataset. Moreover, from Figs. 10 and 11, it can be observed that when the minimum class ranking threshold $\lambda_l = 3$ and the maximum class ranking threshold $\lambda_h = 5$, the proposed method yields the highest segmentation accuracies. Therefore, when $\alpha_0 = 0.01$, $\lambda_l = 3$, and $\lambda_h = 5$, the proposed model can achieve the best segmentation performance.

In addition, an effective UDA semantic segmentation model not only has excellent performance but also needs to meet the computational efficiency requirements. Therefore, we compare the computational efficiency of the proposed method with the baseline [46] and mainstream methods [49]. In the experiment, we use the number of parameters and the floating-point operations per second (FLOPS) as comparison indicators for model efficiency. The experimental results are shown in Table VII. From the table, it can be observed that the number of parameters and FLOPs of our method is slightly higher than the existing mainstream models. While improving the model's performance, it does not result in a great increase in computational complexity, further demonstrating the effectiveness of our method.

V. CONCLUSION

This article proposed an unsupervised remote sensing image semantic segmentation method based on multiscale contrastive domain adaptation, which aims to solve cross-domain remote sensing image differences in resolutions and scenes. The mean teacher model was introduced into the UDA paradigm, and a DCBS was proposed, which is proved to alleviate the class imbalance problems between cross-domain data. The CDMix was developed to reduce the domain gap between the cross-domain data, which can enhance the robustness of the UDA model. Moreover, the MCCL was proposed to learn the domain consistency of the source and target domains. Experimental results on two datasets show that the proposed method can yield superior performance for unsupervised remote sensing image semantic segmentation, which is of great significance to the application of remote sensing intelligent interpretation.

It is demonstrated that the proposed method yields outstanding performance in the UDA remote sensing image semantic segmentation task. However, it can be found in the visualized segmentation map that all UDA methods have a relatively rough segmentation effect on the target boundary. In the future, the segmentation performance of boundary areas under unsupervised conditions will be further improved by enhancing the learning of boundary features.

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